

Original Article

Propofol Versus Remimazolam on Cognitive Function, Hemodynamics, and Oxygenation During One-Lung Ventilation in Older Patients Undergoing Pulmonary Lobectomy: A Randomized Controlled Trial

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Objectives: To investigate the effects of remimazolam on postoperative cognitive function, intraoperative hemodynamics, and oxygenation in older patients undergoing lobectomy.

Design: A prospective, double-blind, randomized, controlled study.

Setting: A university hospital.

Participants: Eighty-four older patients with lung cancer who underwent lobectomy, aged ≥ 65 years.

Interventions: Patients were divided randomly into the remimazolam (group R) and propofol (group P) groups. Group R underwent remimazolam anesthesia induction and maintenance, whereas group P underwent propofol anesthesia induction and maintenance. Cognitive function was assessed with neuropsychological tests 1 day before surgery and 7 days after surgery. The Clock Drawing Test, Verbal Fluency Test (VFT), Digit Symbol Switching Test (DSST), and Auditory Verbal Learning Test-Huashan (AVLT-H) assessed visuospatial ability, language function, attention, and memory, respectively. The systolic blood pressure (SBP), heart rate, mean arterial pressure (MAP), and cardiac index were recorded 5 minutes before induction of anesthesia (T0), 2 minutes after sedation (T1), 5 minutes after intubation with two-lung ventilation (T2), 30 minutes after one-lung ventilation (OLV) (T3), 60 minutes after OLV (T4), and at the end of surgery (T5), and the incidences of hypotension and bradycardia were recorded. The PaO₂, oxygenation index (OI), and intrapulmonary shunt (Qs/Qt) were assessed at T0, T2, T3, T4, and T5. The levels of S-100 β and interleukin 6 were measured by enzyme-linked immunosorbent assay at T0, T5, 24 hours after surgery (T6), and on day 7 after surgery (T7).

Measurements and Main Results: The VFT, DSST, immediate recall AVLT-H, and short-delayed recall AVLT-H scores were significantly higher in group R than in group P on day 7 after surgery ($p < 0.05$). The SBP and MAP at T2 to T5 were significantly higher in group R than in group P, the incidence of hypotension was significantly lower in group R (9.5%) than in group P (35.7%) ($p = 0.004$), and remimazolam significantly reduced the dose of phenylephrine used ($p < 0.05$). The PaO₂ and OI at T4 were significantly higher in group R than in group P, and Qs/Qt was significantly lower in group R than in group P. The levels of S-100 β at T5 were significantly lower in group R than in group P ($p < 0.05$).

Conclusion: The results showed that remimazolam (versus propofol) may lessen the degree of short-term postoperative cognitive dysfunction measured by standard neuropsychological tests, better optimize intraoperative hemodynamics, and lead to improved oxygenation during OLV.

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Key Words: remimazolam; cognitive function; lobectomy; older patients; oxygenation

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PERIOPERATIVE NEUROCOGNITIVE DISORDERS (PND) are a common postoperative central nervous system complication, mainly including postoperative cognitive dysfunction and postoperative delirium.¹ Advanced age is a high-risk factor for PND. It has been reported that the incidence of

postoperative cognitive dysfunction after noncardiac surgery in patients older than 65 at 1 week was 25.8% to 41.4%, and at 3 months, 9.9% to 12.7%.² Meanwhile, the incidence of postoperative cognitive dysfunction in older patients during thoracic surgery is as high as 36.5%.³ This finding may be due to intraoperative one-lung ventilation (OLV) causing a series of pathophysiologic changes (such as ischemia-reperfusion lung injury), resulting in inflammatory responses in peripheral tissues and the central nervous system.^{4,5} Other possible causes of cognitive dysfunction after thoracic surgery include longer anesthetic exposure and intraoperative decline of cerebral oxygen saturation during OLV.^{6,7}

The incidence of PND in older patients undergoing surgery can be decreased by reducing perioperative pain.^{8,9} Also, some studies have reported that anesthetic drugs can improve postoperative cognitive function.¹⁰ Therefore, reasonable selection and proper application of anesthetics are critical for patients.

Propofol has the advantages of rapid induction and recovery, and is one of the most commonly used sedative drugs for lobectomy. Nevertheless, propofol has some unfavorable adverse effects, including pain noted on intravenous injection and dose-related cardiovascular depression.¹¹

Remimazolam is a new type of benzodiazepine with the advantages of a short half-life, quick onset, and recovery.¹² Patients received an initial single intravenous dose of remimazolam, 5.0 mg, over 1 minute. Sedation was maintained by injection of further top-up doses of remimazolam, 2.5 mg (1 mL), not more than 2 minutes apart after assessment of the sedative effect. The overall amount of remimazolam was limited to 5 doses in any 15-minute window. The overall number of midazolam doses was limited to 3 doses in any 12-minute window (as per the midazolam package insert: 1.75 mg initial and/or 1 mg top-up dose for a patient aged <60 years, and 1.0 mg and/or 0.5 mg for those aged 60 years, debilitated, or chronically ill).¹³ The results showed that the restoration of postoperative neuropsychiatric function was better in the remimazolam group than in the midazolam group during colonoscopy. However, the effects of remimazolam on cognitive function after lobectomy in older people have not been studied.

The objective of this study was to evaluate the effects of remimazolam versus propofol on postoperative cognitive function, intraoperative hemodynamics, and oxygenation in older patients undergoing lobectomy.

Methods

Study Design

This study was a prospective, randomized, controlled trial conducted in the authors' hospital from January 2022 to October 2022.

Ethics Approval and Consent to Participate

This study was approved by the ethics committee of the authors' hospital (entry number: IIT2022006) and registered in the Chinese Clinical Trial Center (identifier:

ChiCTR2100050236). This study was conducted in accordance with the principles of the Declaration of Helsinki. The data collection location is the authors' hospital. Written informed consent was obtained from each patient before enrollment.

Patient Inclusion and Exclusion Criteria

The following patients were included in the study. They were aged ≥ 65 years old, had an American Society of Anesthesiologists (ASA) class I to III, and had a body mass index (BMI) of 18-to-28 kg/m². The following patients were excluded from the study: moderate-to-severe reduction of preoperative pulmonary function; anemia before surgery (hemoglobin <90 g/L); significant renal or hepatic insufficiency; allergy to any anesthetic agent; central nervous system or psychiatric disease; low literacy level (illiteracy); hearing or visual disturbances that could affect the reliability of the neuropsychological tests; preoperative memory or cognitive impairment; having undergone previous cognitive function tests; surgery duration <1.5 hours; intraoperative blood transfusion; inability to accomplish the cognitive assessment; reoperation after surgery; and required termination of the trial due to serious adverse reactions during the study.

Randomization and Grouping

Patients were allocated randomly into the remimazolam group (group R) and the propofol group (group P) based on a ratio of 1:1 determined by a computer-generated coding system. A nurse placed the random group sequence number in opaque, sealed envelopes. Another anesthesiologist, who was not involved in the study, randomly removed and opened an envelope about 1 hour before the start of anesthesia to determine the patient grouping, prepare drugs and equipment required for anesthesia, and complete the anesthesia management of the patient. Because of the different colors of propofol and remimazolam, the authors used opaque paper to cover the syringes, infusion channels, and vials of propofol and remimazolam. Patients and researchers who collected data and assessed the outcomes were masked to the drug type and patient grouping.

Protocol

The patients fasted for 8 hours and had no water for 2 hours before surgery. The surgery and anesthesia procedures for all of the patients were performed by the same group of surgeons and by the same group of anesthesiologists, respectively. After entering the operating room, the venous channel was opened, and continuous infusion of lactated Ringer's solution or succinyl gelatin injection (1:1 ratio) was infused at a rate of 8-to-10 mL/kg/h. Radial artery catheterization was performed under local anesthesia, and blood pressure, electrocardiogram, heart rate (HR), finger pulse oxygen saturation (SpO₂), and bispectral index (BIS) were monitored.

In group R, general anesthesia was induced with an initial dose of remimazolam (Jiangsu Hengrui Pharmaceutical Co, Ltd. approval number: 200814AK), 0.3 mg/kg. In group P, general anesthesia was induced with an initial dose of propofol (AstraZeneca Pharmaceutical Co, Ltd. approval number: H2020357), 2.0 mg/kg. Patients received their assigned treatment as an intravenous push by an infusion pump in 1-minute periods. The induction of anesthesia was completed when the patient was fully sedated (defined as BIS <60). If sedation was insufficient (defined as BIS >60) after 1 minute of administration, supplemental doses (remimazolam, 0.1–0.2 mg/kg, or propofol, 0.5 mg/kg) were administered by infusion pump until BIS <60. Sufentanil, 0.5 µg/kg, and cisatracurium, 0.2 mg/kg, were infused intravenously after successful induction. A double-lumen endobronchial tube was used for intubation. The position of the catheter was fixed by the fiberoptic bronchoscopy. The catheter was connected to the anesthesia machine for volume-controlled mechanical ventilation. The mechanical ventilation parameter settings were as follows: during two-lung ventilation, tidal volume of 8 mL/kg, respiratory rate of 12 times/min, inspiratory-to-expiratory ratio = 1:2, and the fraction of inspiration O₂ (FIO₂) of 60%. During OLV, tidal volume of 7 mL/kg, respiratory rate of 14 times/min, inspiratory-to-expiratory ratio = 1:2, FIO₂ of 70% to 80%, and the end-expiratory carbon dioxide partial pressure was maintained in the range of 35-to-45 mmHg.

In group R, anesthesia was maintained with remimazolam (0.6–1.2) mg/kg/h and remifentanyl (0.2) µg/kg/min. In group P, anesthesia was maintained with propofol (2–10) mg/kg/h and remifentanyl (0.2) µg/kg/min. The dose of remimazolam or propofol was adjusted to maintain a BIS range between 40 and 60, as well as intermittent injection of cisatracurium during anesthesia maintenance. Phenylephrine (50 µg per time) was administered if hypotension (defined as systolic blood pressure [SBP] <20% of the basal value lasting at least 1 minute) occurred. Intravenous atropine (0.3 mg per time) was administered if bradycardia (defined as HR <50 beats/min) occurred.

Evaluating Cognitive Function

This study used a set of neuropsychological tests to ensure standardization of the procedure during data collection. Neuropsychological tests were performed 1 day before surgery and 7 days after surgery. The neuropsychological tests consisted of the Mini-Mental State Examination (MMSE), Verbal Fluency Test (VFT), Clock Drawing Test (CDT), Digit Symbol Substitution Test (DSST), and Auditory Verbal Learning Test-Huashan (AVLT-H).

For the CDT, patients were required to score 1 point for drawing a closed circle, 2 points for the correct order and spatial arrangement of all of the numbers on a clock face, and 1 point for the correct position of the hands. The higher the test score, the better the visual-spatial ability.

For the VFT, patients were asked to name as many fruits or vegetables as possible within 1 minute and record the correct

number of these words. The higher the test score, the better the language function.

For the DSST, participants were asked to write Arabic numerals (1–9) that were paired with symbols on a digital array sheet as quickly as possible within 90 seconds. A reference of the digit symbol pairs was provided on the same sheet. The number of digits correctly replaced by the patients within 90 seconds was recorded as the score on the test. This task tests sustained attention, with higher DSST scores indicating better attention.

The AVLT-H has 12 words read aloud by the investigator, and the patient was asked to remember these words. The patient was then asked to recall as many of the 12 words as possible immediately, and the test was repeated 3 times. A short-delayed recall test was conducted 5 minutes later. A long-delayed recall test was conducted 20 minutes later. The immediate recall score was used to assess the patient's ability to learn new information. Short-delayed AVLT-H or long-delayed AVLT-H scores demonstrated the patient's ability to memorize new information. A higher AVLT-H score was associated with better memory ability.

Outcome Assessment

The primary outcome was the assessment data from administered neuropsychological tests (MMSE, CDT, VFT, DSST, and AVLT-H) 1 day before and 7 days after surgery.

The secondary outcomes included the hemodynamic and oxygenation function indicators. Visual analog scale (VAS) scores were recorded 1 day before surgery and day 7 after surgery. Pulse oxygen saturation, SBP, HR, mean arterial pressure, and cardiac index (CI) were recorded at 5 minutes before the induction of anesthesia (T0), 2 minutes after sedation (T1), 5 minutes after intubation with two-lung ventilation (T2), 30 minutes after OLV (T3), 60 minutes after OLV (T4), and at the end of surgery (T5); the incidence of intraoperative hypotension and bradycardia; the dose of phenylephrine used in each group.

The arterial and central venous blood pressures were obtained for blood gas analysis at T0, T2, T3, T4, and T5 to measure PaO₂, oxygenation index (OI), and intrapulmonary shunt (Qs/Qt). The incidence of hypoxemia (defined as intraoperative SpO₂ <90%) and the need for higher FIO₂ (defined as an increase in FIO₂ >80% when intraoperative hypoxemia occurred) were recorded.

Central venous blood was collected at T0, T5, 24 hours after surgery (T6), and on day 7 after surgery (T7). The blood was centrifuged for 10 minutes using a high-speed centrifuge set at 3,000 RPM, serum was extracted and stored at –80°C, and an enzyme-linked immunosorbent assay was performed to detect central nervous system-specific protein (S-100β) and interleukin 6 (IL-6).

General data included age, sex, BMI, ASA classification, education level, forced expiratory volume at 1 second to forced vital capacity ratio (%), and preoperative comorbidities. In contrast, perioperative data included blood loss, operative site, OLV time, operative time, and anesthesia maintenance time.

Statistical Analysis

SPSS 25.0 statistical software (IBM SPSS, Inc, Armonk, NY) was used to analyze the statistics. Continuous data were tested for normality. Normally distributed data were presented as mean $\pm$ SD, Student's *t*-test was used to compare the continuous data, and repeated measurement variance analysis was used to compare multiple time points within groups. Categorical data were presented as absolute values and percentages, and the chi-square or Fisher exact test was used for categorical variables. A $p < 0.05$ was considered statistically significant.

Sample Size Calculation

The sample size was calculated based on a pilot study involving 28 patients (groups R and P, each containing 14 patients). In this pilot study, the mean $\pm$ SD of VFT on day 7 after surgery was 16.9 ± 4.16 scores in group R and 14.3 ± 3.61 scores in group P. Using $\alpha = 0.05$ (two-sided), a target power of 80%, and a 1:1 ratio, the required sample size was 38 patients in each group, as calculated using PASS V11.0 number cruncher statistical system. When considering a loss to follow-up of 10%, the sample size was increased to 42 patients in each group, resulting in a total sample size of 84 patients.

Results

General Characteristics of Patients

The flow chart of this study is shown in Figure 1. A total of 104 patients were included, and 16 patients were excluded

preoperatively. Therefore, 88 patients were included in this study and randomly divided into the R and P groups. One patient in group R was excluded because the surgery time was <1.5 hours, and another patient's surgery was canceled. One patient in group P was excluded because the surgery time was <1.5 hours, and another patient was excluded for an intraoperative blood transfusion. A total of 84 patients ($n = 42/\text{group}$) were finally included and completed the statistical analysis (Fig 1). There were no significant differences in the baseline characteristics of the general data regarding age, sex, ASA classification, BMI, education level, forced expiratory volume at 1 second-to- forced vital capacity ratio, preoperative comorbidities (hypertension or diabetes), intraoperative bleeding, operative site, OLV time, operative time and anesthesia time between the 2 groups ($p > 0.05$) (Table 1).

Cognitive Function

The primary outcomes are reported in Figures 2 and 3. At 1 day before surgery, there were no differences in MMSE, CDT, VFT, DSST, immediate recall, short-delayed recall, or long-delayed recall tests between the 2 groups ($p > 0.05$). The MMSE and VFT scores in group P were significantly lower on day 7 after surgery than on day 1 before surgery ($p < 0.05$). The DSST scores in group R were significantly higher on day 7 after surgery than day 1 before surgery ($p < 0.05$). At day 7 after surgery, there were no significant differences in MMSE or CDT scores between the 2 groups ($p > 0.05$), but VFT and DSST scores were significantly higher in group R than in group P ($p < 0.05$) (Fig 2).

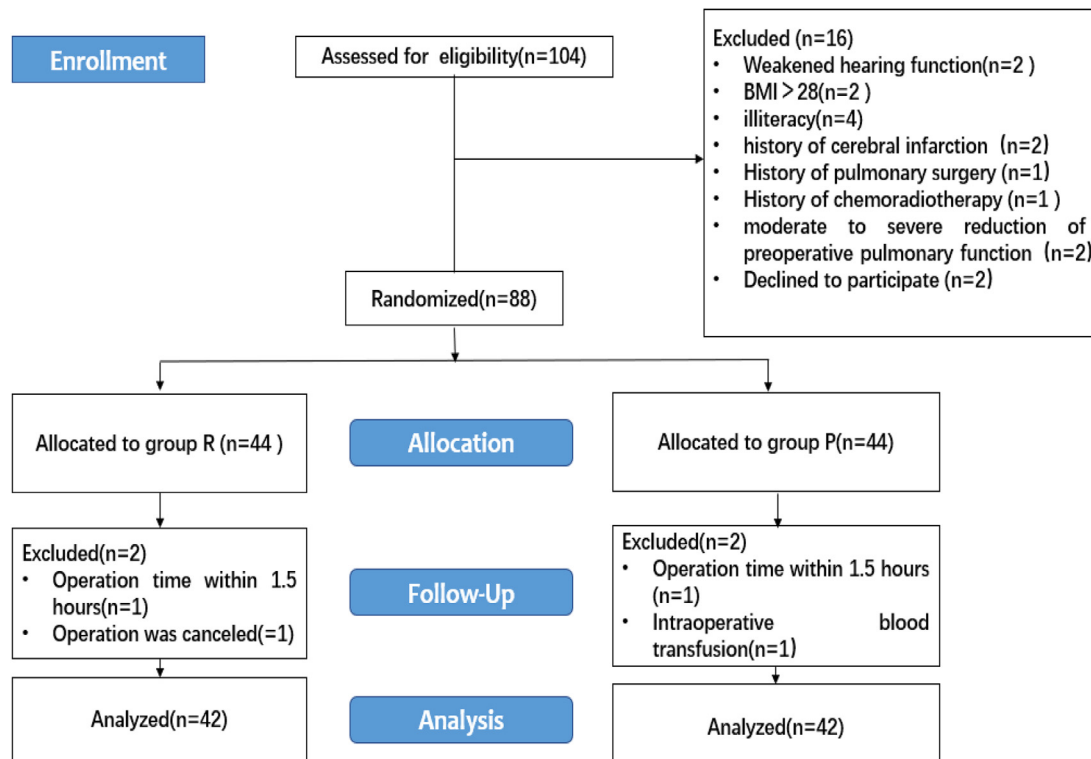


Fig 1. Enrollment flow diagram. BMI, body mass index.

Table 1
Baseline Characteristics of the Patients (n = 42)

Variable	Group R	Group P	p Value
Age, mean $\pm$ SD, y	65.4 $\pm$ 3.9	65.2 $\pm$ 4.4	0.85
Sex			0.82
Male	19	20	
Female	23	22	
ASA, n			0.47
I	14	11	
II	28	31	
BMI, mean $\pm$ SD, kg/m ²	22.4 $\pm$ 2.9	22.3 $\pm$ 2.1	0.97
Education level, n (%), y			
≤ 6	10 (23.8)	8 (19)	
6–9	15 (35.7)	18 (42.9)	
> 9	17 (40.5)	16 (38.1)	
FEV1/FVC ratio, mean $\pm$ SD, %	79.8 $\pm$ 3.4	80.0 $\pm$ 4.8	0.93
History, n (%), yes			
Hypertension	1 (2.4)	1 (2.4)	1
Diabetes	2 (4.8)	1 (2.4)	1
Intraoperative bleeding, mean $\pm$ SD, mL	91.6 $\pm$ 36.9	96.3 $\pm$ 31.4	0.68
Operative site, n			0.23
Right	27	32	
Left	15	10	
OLV time, mean $\pm$ SD, min	122.1 $\pm$ 25.5	128.6 $\pm$ 24.4	0.42
Operational time, mean $\pm$ SD, min	141.3 $\pm$ 31.9	143.9 $\pm$ 22.3	0.77
Anesthesia time, mean $\pm$ SD, min	173.6 $\pm$ 30.8	178.1 $\pm$ 20.0	0.64

Notes: Data are presented as the mean $\pm$ SD or as numbers (%).

Abbreviations: ASA, American Society of Anesthesiologists; BMI, body mass index; FEV₁, forced expiratory volume at 1 second; FVC, forced vital capacity; OLV, one-lung ventilation.

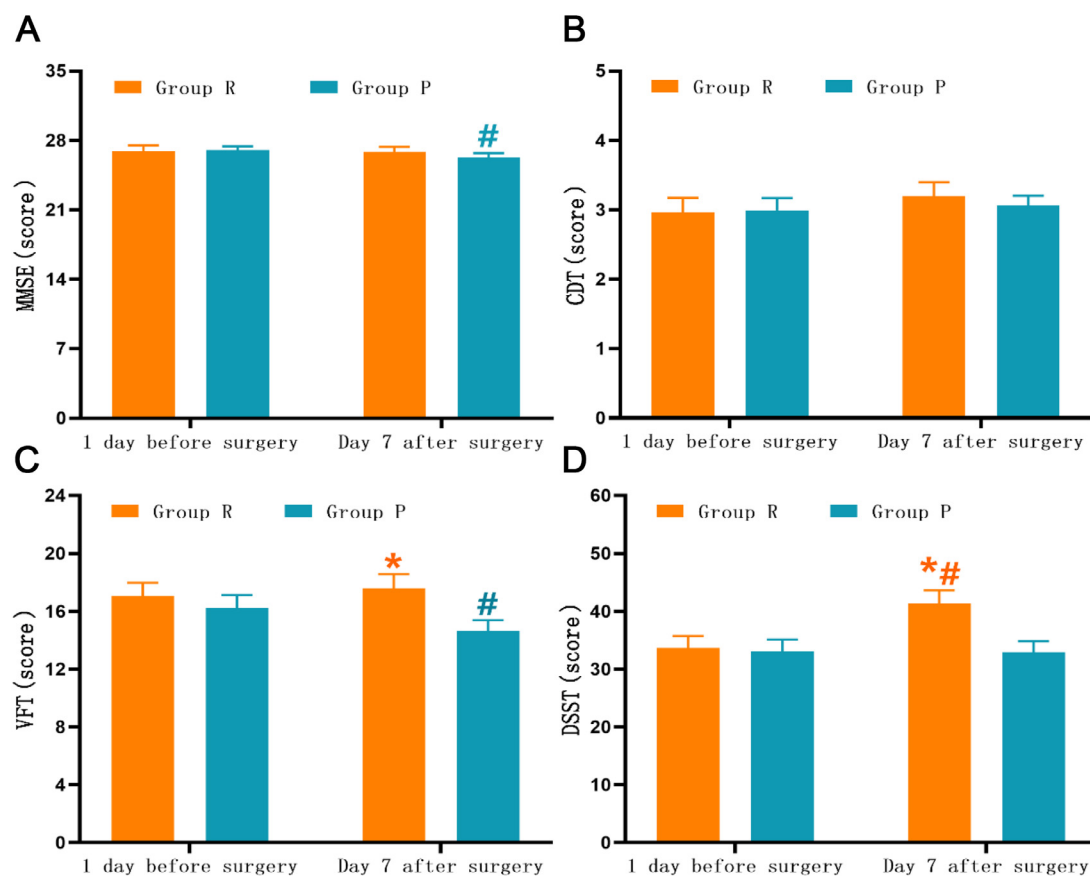


Fig 2. Comparison of Mini-Mental State Examination (A), Clock Drawing Test (B), Verbal Fluency Test (C), and Digit Symbol Substitution Test (D) scores. Data are presented as the mean $\pm$ SEM, n = 42. CDT, Clock Drawing Test; DSST, Digit Symbol Substitution Test; MMSE, Mini-Mental State Examination; VFT, Verbal Fluency Test. #p < 0.05 compared with 1 day before surgery in the group. *p < 0.05 compared with group P.

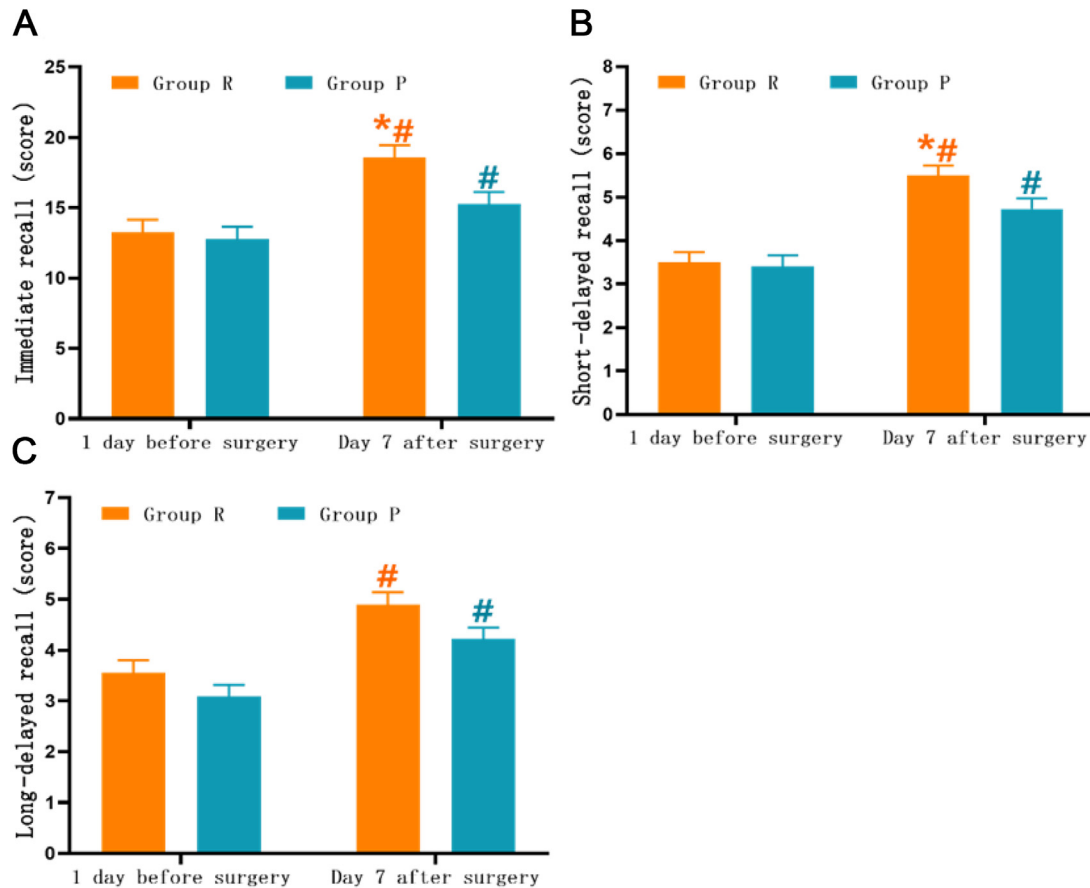


Fig 3. Comparison of scores on the Auditory Verbal Learning Test-Huashan. (A) Immediate recall score. (B) Short-delayed recall score. (C) Long-delayed recall score. Data are presented as the mean $\pm$ SEM, $n = 42$. * $p < 0.05$ compared with group P; # $p < 0.05$ compared with 1 day before surgery in the group.

The immediate recall, short-delayed recall, and long-delayed recall scores were significantly higher in both groups on day 7 after surgery than on day 1 before surgery ($p < 0.05$). At day 7 after surgery, there were no significant differences in long-delayed recall scores between the 2 groups ($p > 0.05$), but the immediate recall and short-delayed recall scores were significantly higher in group R than in group P ($p < 0.05$) (Fig 3).

VAS Scores

There was no difference in VAS scores between the 2 groups on 1 day before surgery ($p > 0.05$). There was no significant difference in VAS scores between the 2 groups on day 7 after surgery ($p > 0.05$). The VAS scores on day 7 after surgery were ≤ 3 , which indicated mild pain that could be tolerated (Table 2).

Hemodynamics

At T0, there were no differences in SBP, MAP, CI, and HR between the 2 groups ($p > 0.05$). From T2 to T5, SBP and MAP were significantly higher in group R than in group P ($p < 0.05$). There were no significant differences in HR or CI between the 2 groups at any timepoint ($p > 0.05$) (Fig 4). In

addition, the incidence of hypotension and the mean dose of phenylephrine given to patients in group R were significantly lower than those in group P ($p < 0.05$). There were no significant differences in the incidence of bradycardia between the 2 groups ($p > 0.05$) (Table 3).

Oxygenation Function

At T0, there were no differences in PaO₂, OI, FIO₂, or SpO₂ between the 2 groups ($p > 0.05$). There were no significant differences in FIO₂ or SpO₂ between the 2 groups at any

Table 2
Comparison of the VAS scores ($n = 42$)

Group	Group R	Group P	p Value
1 day before surgery, n (%)			
0 scores	39 (92.9)	37 (88.1)	0.71
1-3 scores	3 (7.1)	5 (11.9)	0.71
>3 scores	0	0	
Day 7 after surgery, n (%)			
0 scores	4 (9.5)	3 (7.1)	1
1-3 scores	38 (90.5)	39 (92.9)	1
>3 scores	0	0	

Abbreviation: VAS, visual analog scale.

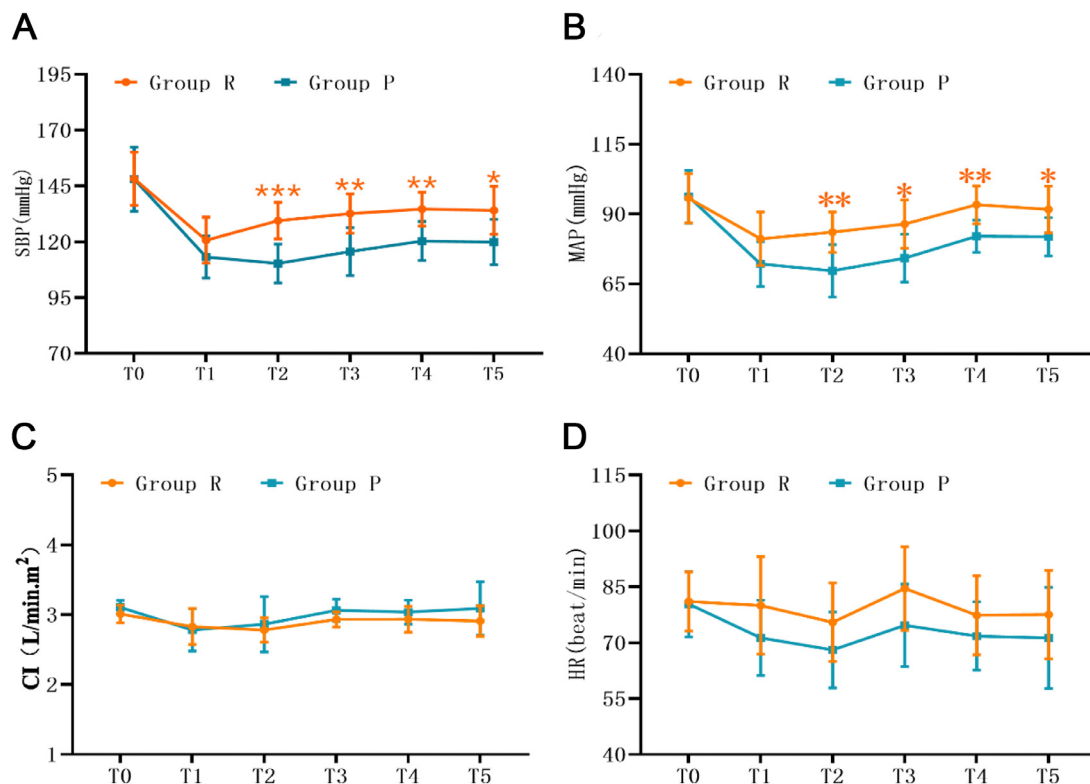


Fig 4. Comparison of hemodynamics. (A) Systolic blood pressure; (B) mean arterial pressure; (C) cardiac index; (D) heart rate. Data are presented as the mean $\pm$ SD, $n = 42$. CI, cardiac index; HR, heart rate; MAP, mean arterial pressure; SBP, systolic blood pressure; T0, 5 minutes before induction of anesthesia; T1, 2 minutes after sedation; T2, 5 minutes after intubation with two-lung ventilation; T3, 30 minutes after one-lung ventilation; T4, 60 minutes after one-lung ventilation; T5, at the end of surgery. * $p < 0.05$ compared with group P.

Table 3
Comparison of the Incidence of Intraoperative Hypotension and Bradycardia ($n = 42$)

Variable	Group R	Group P	P Value
Hypotension, n (%)	4 (9.5)	15 (35.7)	0.004
Phenylephrine, mean $\pm$ SD, μ g	5.96 $\pm$ 22.63	19.05 $\pm$ 31.14	0.030
Bradycardia, n (%)	3 (7.1)	9 (21.4)	0.061

timepoint ($p > 0.05$). At T4, PaO₂ and OI were significantly higher in group R than in group P ($p < 0.05$). At T2, there were no differences in Qs/Qt between the 2 groups ($p > 0.05$). At T4, Qs/Qt was significantly lower in group R than in group P ($p < 0.05$) (Fig 5). There were no significant differences in the incidence of hypoxemia and a higher FIO₂ between the 2 groups ($p > 0.05$) (Table 4).

Inflammatory Response

There were no significant differences in S-100 β or IL-6 between the 2 groups at T0 ($p > 0.05$). From T0 to T7, S-100 β and IL-6 in both groups showed a trend of first increasing and then decreasing. Both S-100 β and IL-6 were significantly higher in both groups at T5 and T6 than at T0 ($p < 0.05$). There was no difference in IL-6 between the 2 groups ($p > 0.05$). However, the level of S-100 β at T5 was significantly

lower in group R than in group P ($p < 0.05$). The S-100 β levels were also lower in group R than in group P at T6 and T7 (Fig 6).

Discussion

In this prospective and controlled study, the results showed that remimazolam (versus propofol) might lessen the degree of short-term postoperative cognitive dysfunction measured by standard neuropsychological tests, better optimize intraoperative hemodynamics, and improve oxygenation during OLV. This finding is important because although remimazolam is used commonly for the sedation of surgical patients, data on cognitive function, hemodynamics, and oxygenation with remimazolam in older patients undergoing lobectomy have not been reported. The findings are worthy of reference for clinicians.

Perioperative neurocognitive disorders have a high incidence, but are an easily overlooked complication characterized by impaired memory, language ability, executive function, and attention and information processing in the days or months after surgery and anesthesia.¹⁴ Older people are the main population for the occurrence of PND. Remimazolam is a new type of ultrashort-acting benzodiazepine that acts on the GABA receptor and quickly produces a sedative-hypnotic effect.¹⁵ The sedative effect of remimazolam is noninferior to propofol for patients undergoing gastroscopy and colonoscopy,

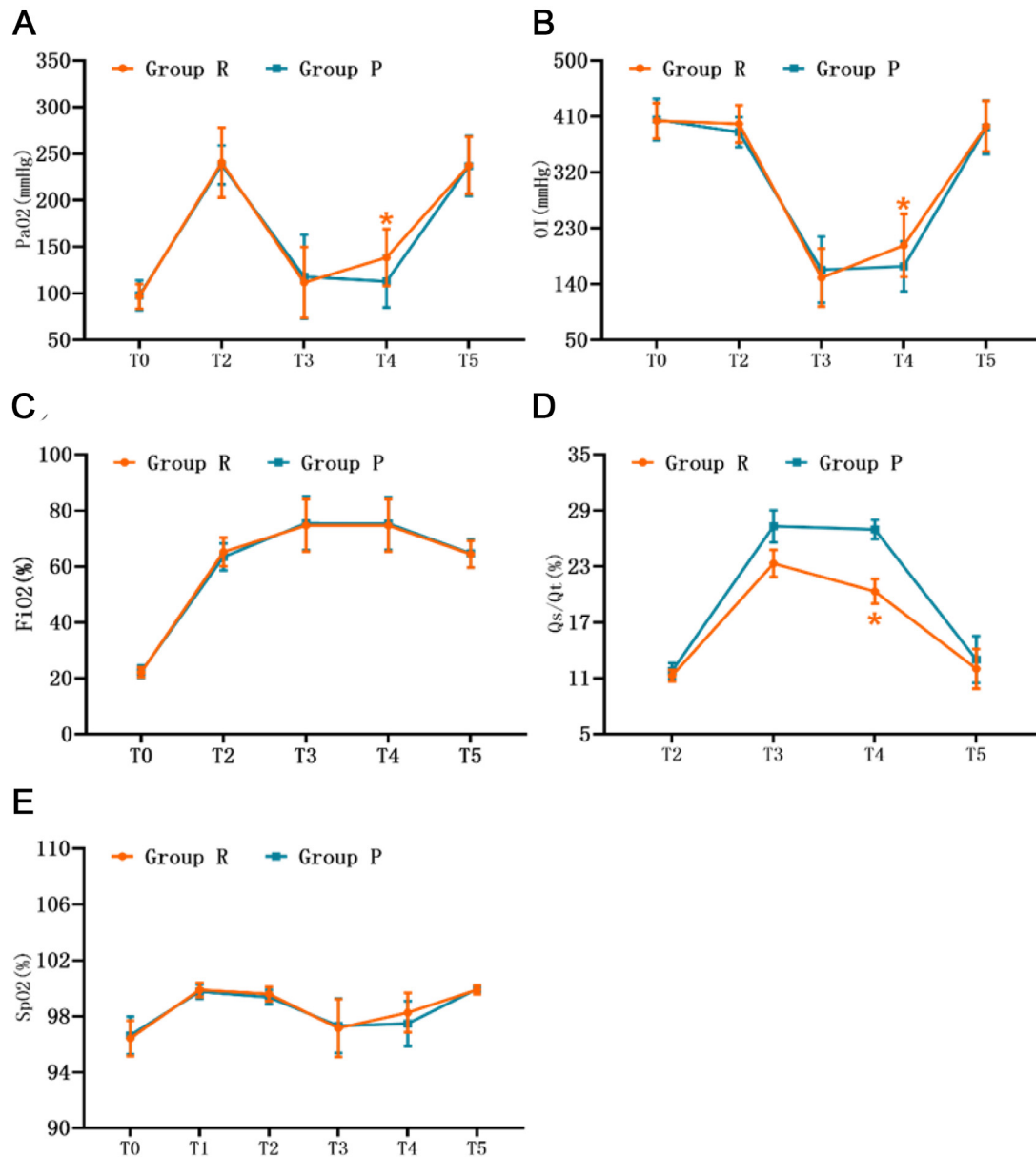


Fig 5. Comparison of oxygenation function. (A) PaO₂. (B) Oxygenation index. (C) Fraction of inspiration O₂. (D) Intrapulmonary shunt. (E) Finger pulse oxygen saturation. Data are presented as the mean $\pm$ SD, n = 42. FiO₂, Fraction of inspiration O₂; OI, oxygenation index; Qs/Qt, intrapulmonary shunt; SPO₂, finger pulse oxygen saturation; T0, 5 minutes before induction of anesthesia; T1, 2 minutes after sedation; T2, 5 minutes after intubation with two-lung ventilation; T3, 30 minutes after one-lung ventilation; T4: 60 minutes after one-lung ventilation; T5, at the end of surgery. *p < 0.05 compared with group P.

Table 4
Comparison of the Incidence of Hypoxemia and Higher FIO₂ (n = 42)

Variable, n (%)	Group R	Group P	p Value
Hypoxemia, %	3 (7.1)	4 (9.5)	1.0
Higher FIO ₂ , %	3 (7.1)	4 (9.5)	1.0

Abbreviation: FIO₂, Fraction of inspiration O₂.

as it is safer and leads to fewer respiratory and circulatory depression events and less incidence of adverse events.^{16,17} The hemodynamic stability of remimazolam is superior to propofol when used as a general anesthetic in heart valve replacement.¹⁸ Remimazolam is used increasingly in surgical

anesthesia, but studies in older lobectomy patients have been rare.

The authors' study was the first to compare short-term assessed cognitive function in patients between 2 groups (remimazolam, propofol) using objective neuropsychological tests (MMSE, CDT, VFT, DSST, and AVLT-H). Their results showed that the DSST and VFT scores on day 7 after surgery were significantly higher in group R than in group P. Their results also showed that the immediate recall and short-delayed recall scores in group R were significantly higher than those in group P on day 7 after surgery, suggesting that remimazolam could improve postoperative attention, language, memory, and recall ability in older patients undergoing lobectomy compared with propofol. It is important that

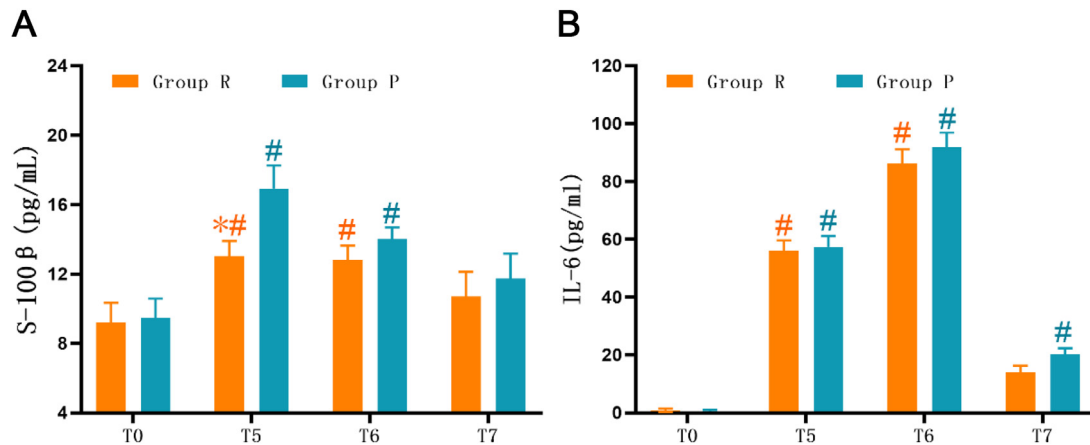


Fig 6. Comparison of levels of central nervous system-specific protein and interleukin 6. (A) Central nervous system-specific protein. (B) Interleukin 6. Data are presented as the mean $\pm$ SEM, $n = 42$. IL-6, interleukin 6; S-100 β , central nervous system-specific protein; T0, 5 minutes before induction of anesthesia; T5, end of the surgery; T6, 24 hours after surgery; T7, on day 7 after surgery. * $p < 0.05$ compared with group P. # $p < 0.05$ compared with T0 in the group.

remimazolam has less interference with postoperative cognitive function and can be used safely for anesthesia and maintenance of lobectomy.

Douglas K. Rex et al. used the HVLT-R as a measure of memory function during colonoscopy at 5 minutes after the patients became fully alert, and found better restoration of neuropsychiatric function in the remimazolam group than in the placebo and midazolam groups,¹³ which was similar to those of the authors' result. Yingjie Tan et al. found that during upper gastrointestinal endoscopy, the scores of immediate recall and short-delayed recall at 5 minutes after recovery to full alertness were significantly lower in the remimazolam group than in the propofol group,¹⁹ which were inconsistent with the authors' results, likely because the time point for assessing postoperative cognitive function in the study was different from their study.

The AVLH test results showed that the scores on day 7 after surgery were significantly higher than the baseline in both groups. It is possible that patients' anxiety and fear led to lower scores before surgery, and learning effects (patients repeatedly using the same tests led to higher scores the second time than the first time) led to higher scores after surgery. In this study, there was no significant difference in VAS scores between the 2 groups, and all postoperative VAS scores were ≤ 3 . The effect of pain on neuropsychological test results was excluded.

During OLV, the alveolar atrophy on the chest-opening side leads to a significant reduction in the area of alveoli involved in gas exchange, resulting in an imbalance in the ventilation-perfusion ratio, an increase in Q_s/Q_t , and an increased incidence of hypoxemia.²⁰ The authors' study results showed that PaO_2 and OI were significantly higher in group R than in group P. It is suggested that remimazolam can improve oxygenation during OLV compared with propofol. Hypoxic pulmonary vasoconstriction (HPV) can reduce V/Q imbalance and Q_s/Q_t .²⁰ However, some drugs, such as inhaled anesthetics and nitroglycerin, inhibit HPV.²⁰ The authors' study results showed that Q_s/Q_t was significantly lower in group R than in group P. They speculate that remimazolam (versus propofol)

has less effect on the MAP and SBP, resulting in better maintenance of HPV.

At the same time, the authors' results also showed that the SBP and MAP during most of the intraoperative period were significantly higher in group R than in group P, and the incidence of hypotension was significantly lower in group R than in group P. Because of the immediate treatment of hypotension, the dosages of phenylephrine in group R were correspondingly reduced compared with those in group P. It indicated that remimazolam had better hemodynamic stability than propofol in older patients undergoing lobectomy, which was consistent with the previous study.^{18,21,22}

The S-100 β is a cytoplasmic calcium-binding protein,²³ and is synthesized mainly by astrocytes.²⁴ It is often used as an indicator of brain damage. Some studies have suggested that elevated S-100 β levels might be associated with cognitive dysfunction.^{25,26} The results of this study showed that the level of S-100 β at T5 was significantly lower in group R than in group P. The S-100 β levels were also lower in group R than in group P at T6 and T7. It was suggested that there was less perioperative neuronal damage in group R than in group P. The authors also observed that the day 7 score in group P was near its preoperative baseline; group R was above its own baseline, which may have been related to the decrease in S-100 β with remimazolam (versus propofol). To find possible reasons why remimazolam may lessen postoperative cognitive dysfunction, the authors' study also tested IL-6 levels. It has been found that the central inflammatory response can damage neuronal cells, leading to postoperative cognitive dysfunction.²⁷ Ischemia-reperfusion lung injury during OLV can cause the release of proinflammatory cytokines (such as IL-6), causing peripheral and central inflammatory responses.^{4,5} However, the results of this study showed that the level of IL-6 in group R was not significantly different from that in group P, suggesting that remimazolam may lessen postoperative cognitive dysfunction, likely not by inhibiting ischemia-reperfusion lung injury.

This study had some limitations that need to be discussed. First, this was a single-center study; more multicenter studies

are needed. Second, this study compared the effects of remimazolam and propofol on short-term cognitive function (on day 7 after surgery) in older patients undergoing lobectomy, and further studies are needed to compare the effects on long-term cognitive function (on day 7 after surgery or longer). Third, using inhaled anesthetics (eg, sevoflurane) alone in lobectomy is rare in China. It is necessary to set up an inhaled anesthesia group in future studies.

Conclusion

The results showed that remimazolam (versus propofol) may lessen the degree of short-term postoperative cognitive dysfunction measured by standard neuropsychological tests, better optimize intraoperative hemodynamics, and improve oxygenation during OLV.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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